

How Predictive Modeling Functions Work in Tableau

Applies to: Tableau Cloud, Tableau Desktop, Tableau Public, Tableau Server

You can already add [trend lines](#) and [forecasts](#) to a visualization, but now you can go beyond this, using the power of a statistical engine to build a model that understands how your data is distributed around such a trend line, or line of best fit. Previously, users had to integrate Tableau with R and Python in order to perform advanced statistical calculations and visualize them in Tableau. Now, you can use the predictive modeling functions to make predictions from your data by including them in a table calculation. For more information about table calculations, see [Transform Values with Table Calculations](#).

With these predictive modeling functions, you can select targets and predictors by updating the variables and visualizing multiple models with different combinations of predictors. The data can be filtered, aggregated, and transformed at any level of detail, and the model—and thus the prediction—will automatically recalculate to match your data.

For a detailed example that shows how to create prediction calculations using these functions, see [Example - Explore Female Life Expectancy with Predictive Modeling Functions](#).

Predictive modeling functions available in Tableau

MODEL_PERCENTILE

Syntax	<pre>MODEL_PERCENTILE(model_specification (optional), target_expression, predictor_expression(s))</pre>
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Definition	Returns the probability (between 0 and 1) of the expected value being less than or equal to the observed mark, defined by the target expression and other predictors. This is the Posterior Predictive Distribution Function, also known as the Cumulative Distribution Function (CDF).
Example	<pre>MODEL_PERCENTILE(SUM([Sales]), COUNT([Orders]))</pre>

MODEL_QUANTILE

Syntax	<pre>MODEL_QUANTILE(model_specification (optional), quantile, target_expression, predictor_expression(s))</pre>
Definition	Returns a target numeric value within the probable range defined by the target expression and other predictors, at a specified quantile. This is the Posterior Predictive Quantile.
Example	<pre>MODEL_QUANTILE(0.5, SUM([Sales]), COUNT([Orders]))</pre>

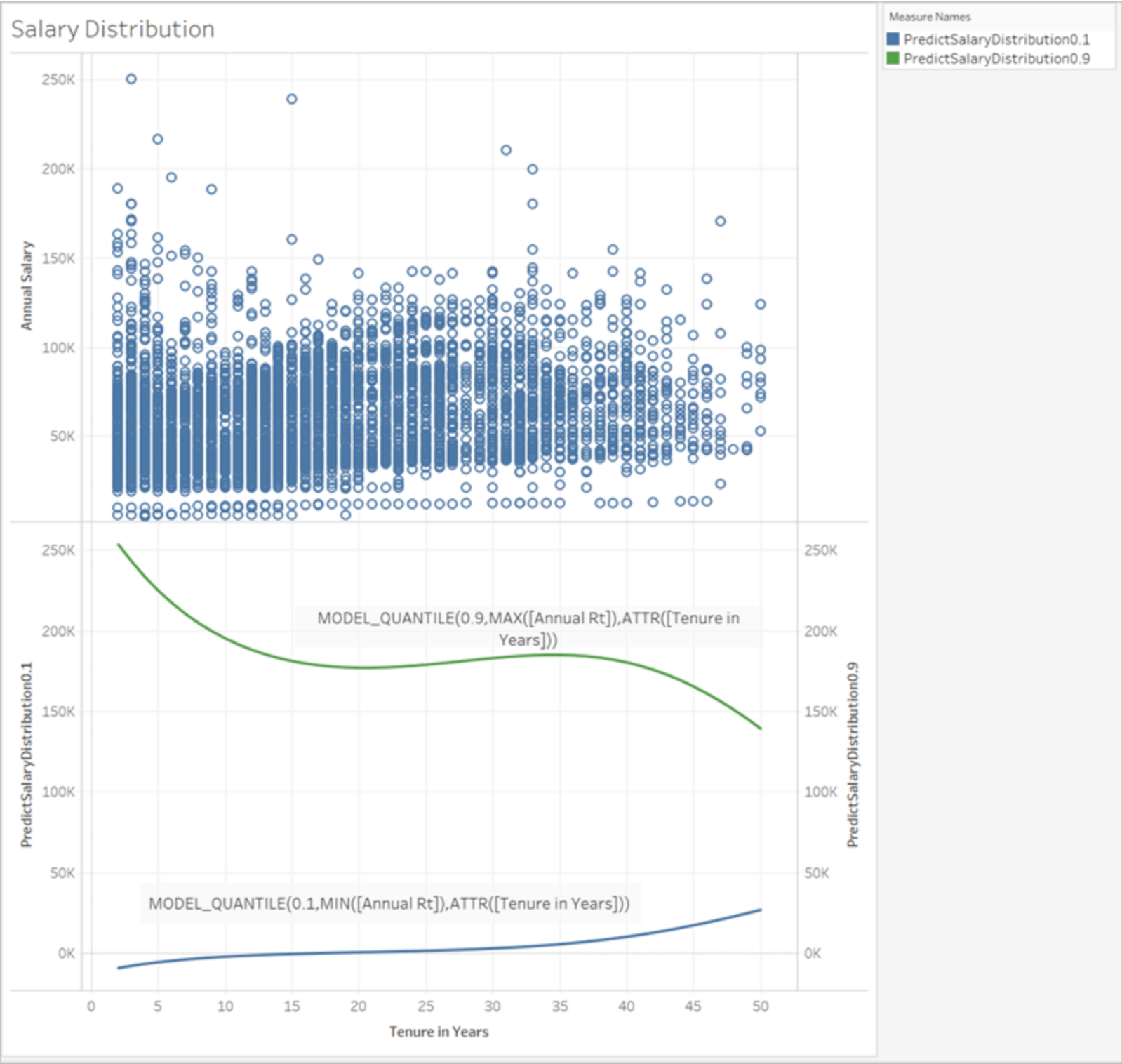
The power of predictive modeling functions

Let’s look at an example using salary data, starting with MODEL_QUANTILE.

In the example below, MODEL_QUANTILE has been used to display the 10th percentile and the 90th percentile of the expected distribution for the same data set. Based on the existing data and using a linear regression model, the statistical engine has determined that there is a 90% probability that the maximum salary for each tenure will be below the green line, and a 10% probability that the minimum salary for each tenure will be below the blue line.

In other words, with the quantile set at 0.9, the model predicts all salaries will fall at or below the green line 90% of the time. The blue line is set at .1 or 10%, so only 10% of salaries will fall at or below the blue line, with the inverse of that (90%) above the blue line.

Effectively, this gives us a band that we can predict 80% of potentially generated future points or unobserved data will fall inside.



Next, let's see how MODEL_PERCENTILE, the inverse of MODEL_QUANTILE, can help us understand the data further.

You can identify the outliers of the data set by using the MODEL_PERCENTILE function. MODEL_PERCENTILE tells you, as a percentile, where the observed mark falls within a range of probable values for each mark. If the percentile is very close to 0.5, the value observed is very close to the median value predicted. If the percentile is close to 0 or 1, the value observed is at the lower or upper boundaries of the model range and is relatively unexpected.

Below, we have applied MODEL_PERCENTILE as color to the salary visualization in the top half of the image, to help us understand which values are most expected.



Syntax of predictive modeling functions in detail

What is MODEL_QUANTILE?

MODEL_QUANTILE calculates the posterior predictive quantile, or the expected value at a specified quantile.

- **Quantile:** The first argument is a number between 0 and 1, indicating what quantile should be predicted. For example, 0.5 specifies that the median will be predicted.
- **Target expression:** The second argument is the measure to predict or “target.”
- **Predictor expression(s):** The third argument is the predictor used to make the prediction. Predictors can be dimensions, measures, or both.

The result is a number within the probable range.

You can use MODEL_QUANTILE to generate a confidence interval, missing values such as future dates, or to generate categories that don't exist in your underlying data set.

What is MODEL_PERCENTILE?

MODEL_PERCENTILE calculates the posterior predictive distribution function, also known as the Cumulative Distribution Function (CDF). This calculates the quantile of a particular value between 0 and 1, the inverse of MODEL_QUANTILE.

- **Target expression:** The first argument is the measure to target, identifying which values to assess.
- **Predictor expression(s):** The second argument is the predictor used to make the prediction.
- Additional arguments are optional and are included to control the prediction.

Notice that the calculation syntax is similar, with `MODEL_QUANTILE` having the extra argument of a defined quantile.

The result is the probability of the expected value being less than or equal to the observed value expressed in the mark.

You can use `MODEL_PERCENTILE` to surface correlations and relationships within your database. If `MODEL_PERCENTILE` returns a value close to 0.5, the observed mark is near the median of the range of predicted values, given the other predictors that you've selected. If `MODEL_PERCENTILE` returns a value close to 0 or to 1, the observed mark is near the lower or upper range of what the model expects, given the other predictors that you've selected.

Note: Dimensions used as predictors can be *ordered* or *unordered*. An ordered dimension is any whose values can be sequenced, such as `MONTH`. An unordered dimension is any whose values don't have an inherent sequence, such as gender or color. This distinction matters when using Gaussian process regression. For more information on this model and others, see [Choosing a Predictive Model](#).

For advanced users, there are two other optional arguments you can include to control the prediction. For more information, see [Regularization and Augmentation in Predictive Modeling](#).

What's being computed?

The input used to build the model is a matrix where each mark is a row, and the columns are the target expression and predictor expressions evaluated for each mark. Whatever row is specified on the viz is what defines the row for the data set being computed by the statistical engine.

Let's look at the example below, where rows (and thus marks) are defined by job titles, and columns are the target expression `MEDIAN([Annual Rt])`. These are followed by the optional additional predictors `MEDIAN([Tenure in Months (Measure)])` and `ATTR([Department Generic (group)])`.

Jobtitle	Actual Median Annual Rt	Median Tenure in Months (Measure)	Department
911 Lead Operator	60,425	287	Fire Department
911 Operator	53,134	141	Fire Department
911 Operator Supervisor	66,537	423	Fire Department
Account Executive	60,700	35	M-R
Account Executive Supervisor	73,900	70	M-R
Accountant I	52,268	75	*
Accountant II	61,451	102	*
Accountant Supervisor	71,300	225	*
Accounting Assistant I	32,003	113	*
Accounting Assistant II	34,011	61	*
Accounting Assistant III	46,758	206	*
Accounting Asst II	32,271	58	M-R
Accounting Asst III	41,810	60	M-R
Accounting Manager	88,100	123	*
Accounting Systems Admin	68,136	137	Finance

Note: Several job titles have an asterisk listed under Department. This is because those job titles exist in several departments but Tableau treats all of them as though they are in the same department. In addition, the statistical engine doesn’t consider the number of records, and treats each mark as equally likely. This is because Tableau can’t perform table calculations on unaggregated data, and can’t aggregate the results of table calculations. For more information on aggregation, see [Data Aggregation in Tableau](#).

For more information on the data used to build a model and generate predictions, see [Compute Using and Data Partitioning in Predictive Modeling](#).

What models are supported?

Predictive modeling functions support linear regression, regularized linear regression, and Gaussian process regression. These models support different use cases and prediction types, as well as have different limitations. For more information, see [Choosing a Predictive Model](#).

Choosing your predictors

A predictor can be any field in your data source—measure or dimension—including calculated fields.

Note: If using a dimension as a predictor, you must use a ATTR wrapper (e.g. ATTR([State]) rather than just [State]). This is because dimensions will generally only work if they are in the level of detail in the visualization either directly, or above it in a hierarchy.

For example, suppose you have a data set that includes the fields [City], [State], and [Region], where multiple [City] records exist within a [State], and multiple [State] records within a [Region].

In a visualization that uses [State] as a mark, the predictor ATTR([State]) or ATTR([Region]) will both work as predictors. However, the predictor ATTR([City]) will revert to *, since multiple cities exist within a visualized state and so cannot be used as predictors. In other words, including a predictor that's at a lower level of detail than the viz adds no value to the predictions. In most cases, a predictor at a lower level of detail than the viz will evaluate to * and so will all be treated identically.

However, if the same data set is used to generate a viz that uses [City] as a mark, ATTR([City]), ATTR([State]), and ATTR([Region]) can all be used successfully as predictors. For more information about the use of ATTR functions, see [When to Use the Attribute \(ATTR\) Function](#).

Dimensions and measures do not need to be visualized (in the view or viz) in order to be included as predictors. For more detailed guidance, see [Choosing Predictors](#).

Recommendations

Prediction calculations are best used as follows:

- To predict values for individual records, where each mark in your viz represents a discrete entity such as a product, sale, person, etc., rather than aggregated data. This is because Tableau regards each mark as equally likely, even if one mark is made up of 100 records and the other marks are made up of one record each. The statistical engine does not weigh marks based on the number of records that make it up.
- To predict values for aggregated target expressions using SUM and COUNT.

Note: MODEL_QUANTILE and MODEL_PERCENTILE aren't recommended to predict values for aggregated target_expressions using AVG, MEDIAN, MIN, and MAX.

Limitations

- You must use a calculated field to extend a time series into the future. For more information, see [Predicting the Future](#).
- Predictors should be at the same or higher level of detail than the view. That is, if your view aggregates by state, you should use state or region as a predictor, but not city. For more information, see [Choosing Predictors](#).

When will prediction calculations break?

Regardless of the model you're using, you must have at least three data points within each partition for the model to return a response.

If you have specified Gaussian process regression as your model, it can be used in predictive calculations with one **ordered** dimensional predictor and any number of **unordered** dimensional predictors. Measures aren't supported as predictors in Gaussian process regression calculations, but can be used in linear and regularized linear regression calculations. For more information on model selection, see [Choosing a Predictive Model](#).

If your calculation used ATTR[State] as a predictor and your viz also included State as a mark but no other field at a lower level of detail, such as City, then you would return an error. To prevent this, simply ensure there is not a one-to-one relationship between marks and predictor categories.

For more information on these and other prediction problems, see [Resolve Errors in Predictive Modeling Functions](#).

FAQ

What about marks in multiple predictor-groups?

If a row is aggregated from data that exists within multiple predictor-groups, the value of the ATTR function is a special value of multiple values. For instance, all cities that exist in multiple states will have the same predicted value (unless there are other predictors that are distinct). When selecting your predictors, it's best to use predictors that are at the same level of detail as the viz or or higher. Again, for more information on ATTR functions, see [When to Use the Attribute \(ATTR\) Function](#).

What if ATTR Aggregation returns a * value?

* is treated as a distinct value. If ATTR returns * for all marks, then you, in essence, have a predictor with a constant value, which will be ignored. This is the same as not including that predictor at all.

If ATTR returns * for some but not all marks, then it will be treated as a category where all the * values are considered to be the same. This scenario is identical to the above scenario, in which marks exist in multiple predictor-groups.

What about the "Compute Using" table calculation menu options?

This functions identically to Compute Using in other table calculations. For more information, see [Compute Using and Data Partitioning in Predictive Modeling](#).

Why am I getting an error?

There are several reasons why you might be encountering an error when using predictive modeling functions. For detailed troubleshooting steps, see [Resolve Errors in Predictive Modeling Functions](#).